

wireless radio on a headset worn by the latter in such situations as disputed catches and boundaries, run out decisions and so on.

Hawk Eye technology

This includes a computer and a camera system to trace a ball's trajectory. It is used in international cricket, tennis, soccer and many other sports. It uses six or more computer-linked TV cameras placed around the field. Views from different TVs are then combined to produce an accurate 3D representation of the path of the ball. The technology then traces the ball's trajectory and sends it to a VR machine.

High-speed cameras are also used to take 600 frames a second on, for example, goal-line of the football.

This technology allows broadcasters to showcase other features of games like cricket, such as comparing bowlers' speeds, spin, swing, line and length. In the case of leg-before wicket (LBW) decisions, it can project the likely path of the ball forward, through the batsman's legs, to see if it would have hit the wicket. Images captured through high-speed cameras focused on the ball can also measure ball speed (revolutions per minute), showing how fast the ball was spinning after release.

Goal detection system

This is a radio-based system that uses low-frequency magnetic fields

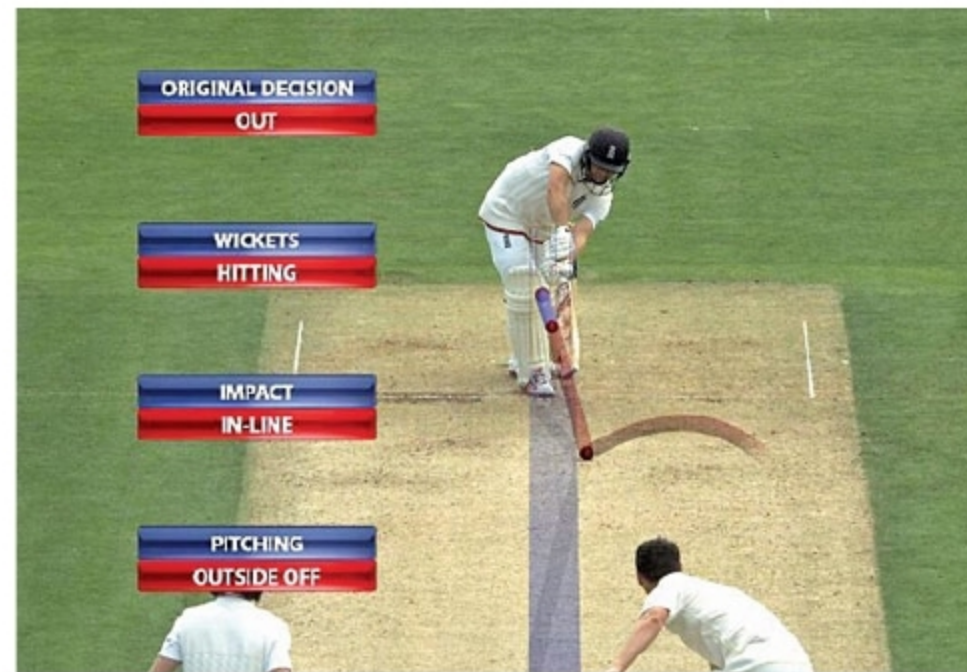
to determine whether or not the whole ball has passed over the goal line in a game of football. There are two magnetic fields: one in the goal area, and the other created in and around the ball. Data is processed and then transmitted wirelessly to the referee's wrist-watch. A message is displayed in real time on the watch.

Goal-line technology is used to determine if a ball has crossed the goal line, to support the referee. A smart ball loaded with a sensor suspended inside can also be used. It utilises a network of receivers around the field designed to track the ball's precise position in real time. In football, goal-line sensors were introduced as part of goal-line assessment to give a definitive decision on whether the ball had crossed the line.

Snick-o-meter

Used for cricket, this technology comprises a sensitive microphone embedded in one of the stumps and connected to an oscilloscope that displays sound waves. When the ball hits the bat, the oscilloscope trace picks up the sound. The trace is shown in a slow-motion video to the TV audience.

From the shape and timing of the sound wave, the viewer can determine whether the sound came from the bat hitting the ball or from some other object like a leg-pad. A short sharp sound is associated with a bat on the ball, while a bat hitting the pad or the ground produces a flatter sound wave. The umpire does not get the



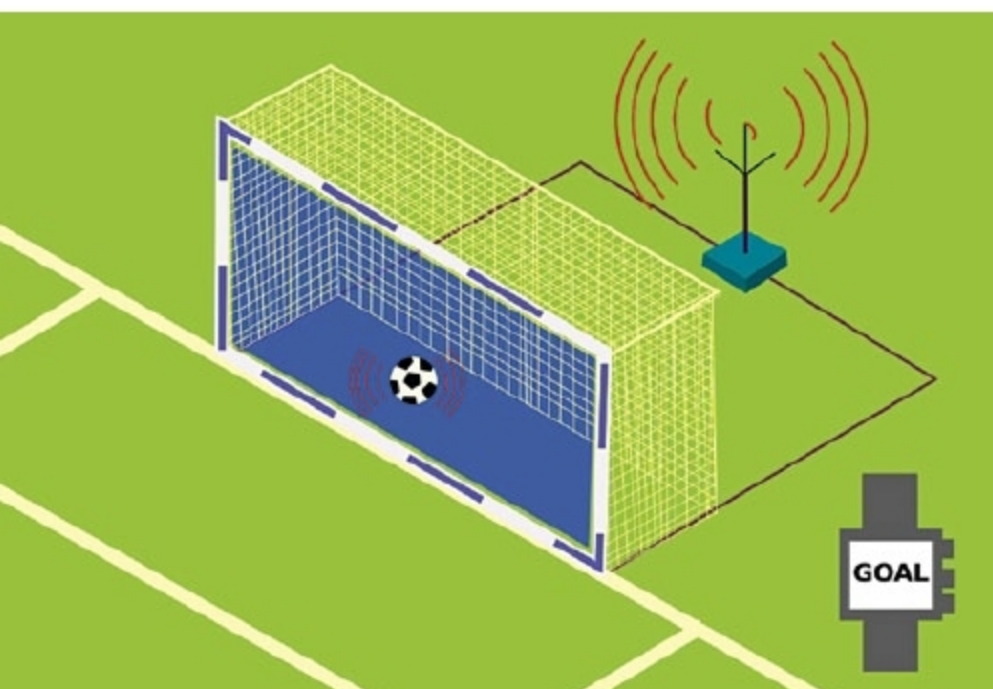
Hawk Eye technology being used in cricket (Credit: www.hawkeyeinnovations.com)

benefit of seeing the snick-o-meter, and has to rely on his/her judgement or refer to the third umpire in case of any doubt.

Technologies to track the health and performance of athletes

Game analysis generates statistical descriptions of gameplay. This allows coaches and players to identify the strengths and weaknesses in their own and opponent's game strategies. Field-based assessment and analysis help athletes of all sports to achieve their full potential for optimal sports speed. Laser-based Fast technology is one such highly accurate speed and acceleration testing and training system that calculates a runner's force, acceleration, speed and time.

Wearable devices that can monitor heart rate are integrated with tracking technologies such as GPS, accelerometer and gyroscope sensors to describe athletes' movements and physical demands. These can assess the number of collisions and jumps that occur during a match, knowing exactly where and how the players move on the field. Athletic trainers use such data to plan training sessions and use technologies to provide measurable training routines, customised for each athlete. Team doctors can utilise heart rate and sensors data



Goal-line technology being used in football (Credit: Wikipedia)